Computer Science & IT



C Programming

Array & Pointers

Lecture No. 02



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Recap of Previous Lecture



- Topic — Pointers
- Topic — Call by reference
- Topic
- Topic
- Topic

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Call by reference ← Swap

Array ← pointer + pointer Arithmetic

function Returning pointer

Slide



Call by reference



```
#include<stdio.h>
void foo(int*);
int main(){
    int a=20
    foo(&a);
    printf("%d",a);
    return 0;
}
```

parameter Integer pointer

```
void foo(int *ptr){
    *ptr = *ptr+20;
}
```

a [2040] 1000

ptr [1000]

40

Slide



function Returns pointer

prototype : int foo(int)

foo is a function which take 1 integer argument & return an integer

prototype : int *bar(int)

bar is a function taking one integer as argument return an integer pointer

Slide

function Returns pointer



problem

```
# include<stdio.h>
int * bar(int);
int main(){
   int a=20;
   int *ptr;
   ptr = bar(a);
   printf("%d", *ptr);
}
```

i | 20 | 1000

ptr = | 1000 |

why : Later

```
int* bar (int b){
                20
int i=20;
b= i+20;   b=40
return &i;
}
```

Local variable

Slide



function Returns pointer

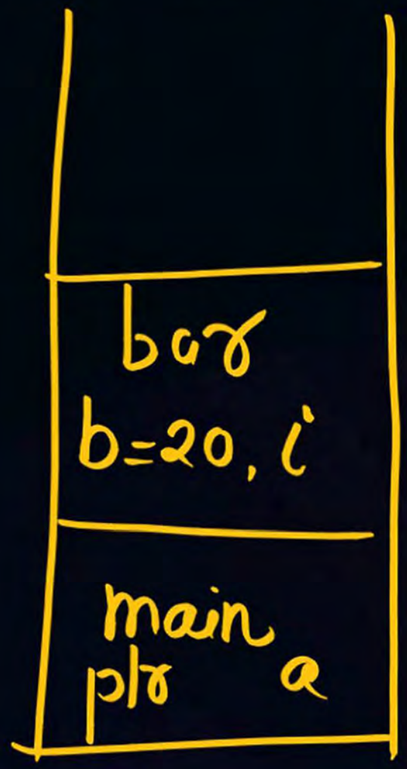


Dangling pointer is a pointer holding Address of a variable which does not exists.

Slide



Function Returns pointer.



Stack



When bar function terminates Local variable destroy. ptr is having an address which deallocated (destroyed, does not exist.)

Dangling pointer

Slide

float :- IEEE-754 } float & Double

floating point representation

Imp · COA



Call by reference

#Q What does the following program print?

```
#include<stdio.h>
void f(int *p, int *q) {
    p=q;
    *p=2;
}
int i=0, j=1;
int main() {
    f(&i, &j);
    printf("%d %d\n", i,j);
    return 0;
}
```

(A) 2,2
(B) 2,1
(C) 0,1
(D) 0,2

p 1000 2000

q 2000

i 0 1000 j 2 2000

Call by reference

p − 1000
q = 2000

p=q , p is update

*p = 2

a=2 a=b
b=4 a=4

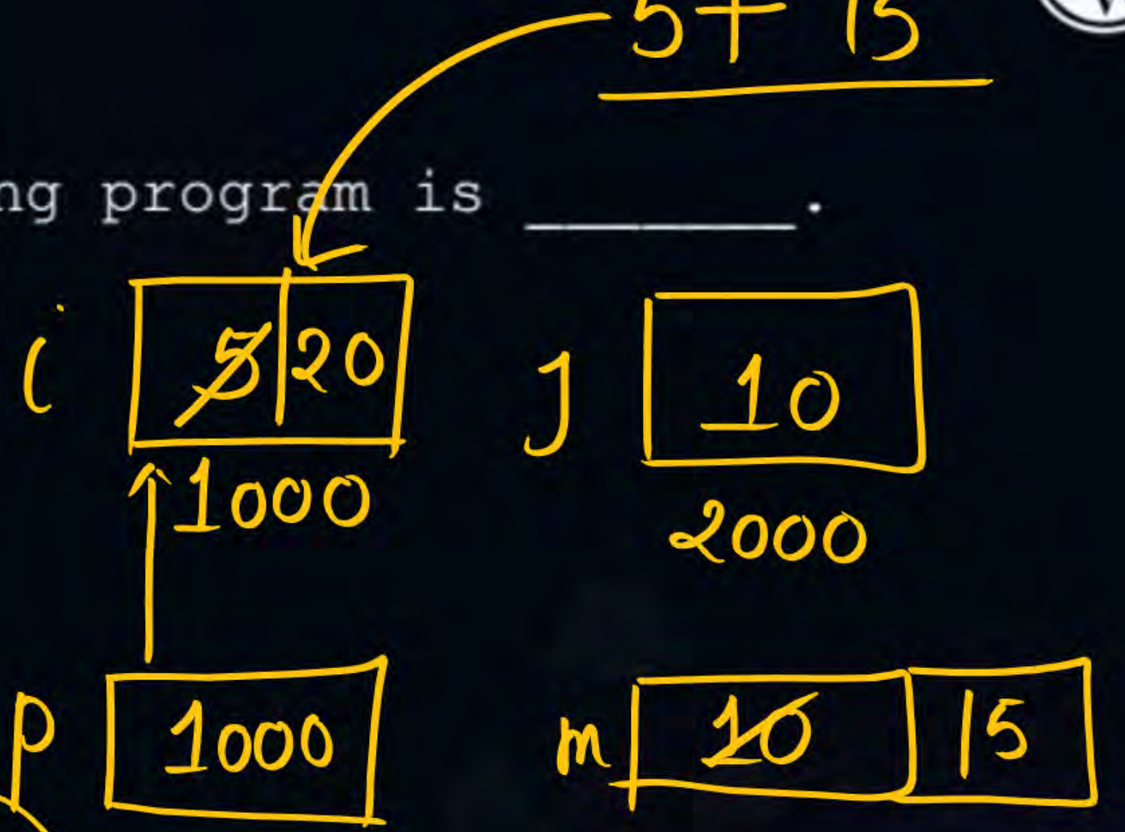
0,2

Slide

GATE 2016

The value printed by the following program is _____.

```
void f (int * p, int m) {
    m = m + 5;
   *p = *p + m;
    return;
}
void main () {
 int i=5, j=10;
 f (&i, j);
 printf ("%d", i+j);
}
```



Slide

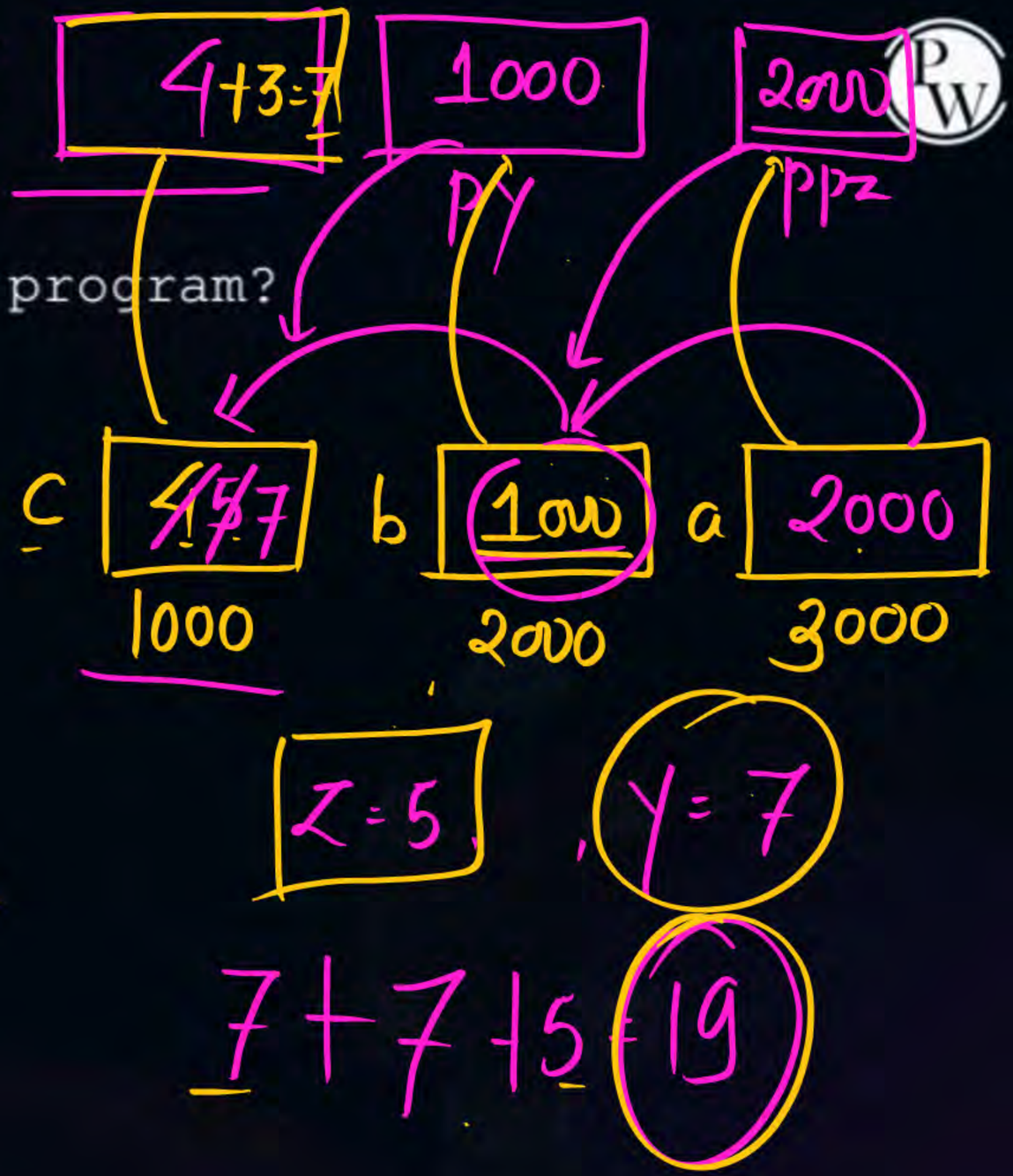
GATE 2008

What is printed by the following C program?

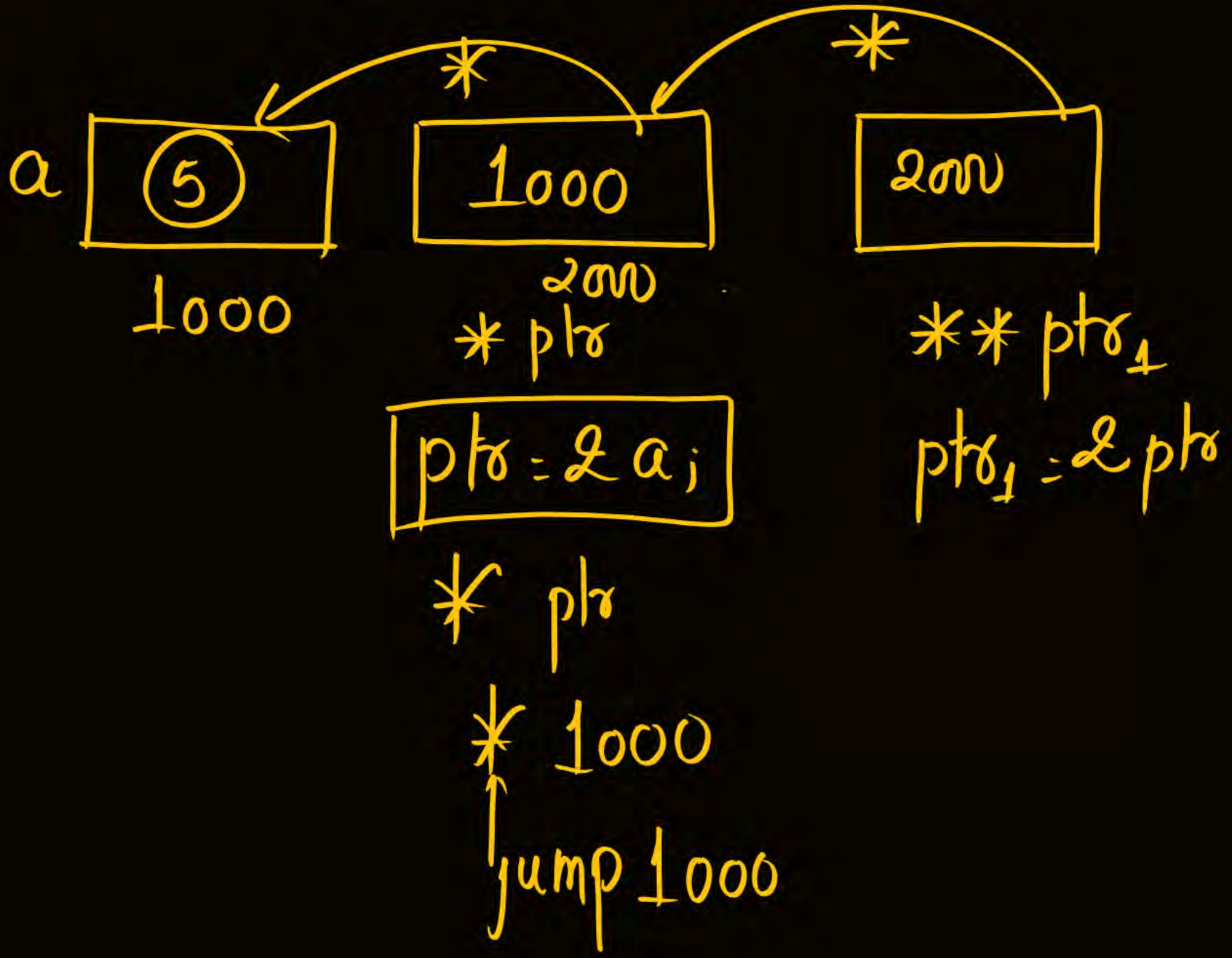
```
int f(int x, int *py, int **ppz){
    int y, z;
    **ppz += 1; z = **ppz;
    *py += 2; y = *py;
    x += 3;
    return x+y+z;
}
void main(){
    int c, *b, **a;
    c = 4; b = &c; a = &b;
    printf("%d", f(c, b, a));
}
```

A. 18
B. 19
C. 21
D. 22

4 1000 2000



Slide


a
5
1000
1000
2000
2000
* ptr
ptr = &a;
** ptr1
ptr1 = &ptr
* ptr
* 1000
jump 1000



2 mins Summary

Topic

Topic

Topic

Topic

Topic

Call by reference

function Returning pointer

Dangling pointer

problem Solve

Slide

THANK - YOU

